# Diego **Zamboni**

Organizational Leader · Security Expert · Computer Scientist

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## Key skills

| | |
|---|---|
| **Leadership** | 32 years of multidisciplinary team and project leadership experience; IT Enterprise Architecture; Scaled Agile Framework (SAFe) experience. |
| **Communications** | Excellent written and spoken communication skills, extensive public speaking, writing and teaching experience. |
| **Technology** | Broad and deep IT expertise, including cloud computing, computer security, operating systems, networking, configuration management, software & services development, programming languages, etc. |
| **Cloud Computing** | AWS architecture, infrastructure, development and security. Holder of 7 AWS certifications. |
| **Computer Security** | Enterprise security architecture; virtualization and cloud computing security architecture and implementation; risk management and compliance; intrusion detection and prevention; software security and secure software development. |
| **Research** | Ph.D. in Computer Science, 9 years of experience at IBM Research. |
| **Languages** | Spanish (native), English (100%), German (B2 level). |

## Professional highlights

- Management and leadership, IT security, cloud computing
  - Currently manage security architecture at the Stellantis *Virtual Engineering Workbench* project. Worked with CISO and business stakeholders to define governance, establish security best practices and drive risk analysis, threat modeling and mitigation.
  - Established scalable and durable mechanisms to enable partners to work securely in the Volkswagen Digital Production Platform (DPP) program.
  - Managed security architecture, risk management, data governance and compliance (ISO27001, ISAE3402/3000, etc.) for Swisscom's Cloud platforms.
  - Established and led the Swisscom IT Clouds security community of practice.
  - Established and led the *Health and State Management* team at Swisscom to design, implement and operate a framework for scalable monitoring, logging and alerting for Swisscom's Cloud platforms.
  - Established and led the first computer security organization at UNAM, which has grown into the university's Information Security Coordination (UNAM-CERT).
  - Managed IT security customer relationships at HP Enterprise Services, including overseeing the activities of operational and engineering teams, risk and compliance management, requirements discussion and reporting.
- Research, architecture and design
  - Designed the *Orchard* monitoring framework for Swisscom's *Application Cloud* platform, and led the team that implemented it and brought it into production.
  - Designed and implemented the *Billy Goat* malware capture and analysis system at IBM.
- Communications and community
  - Author of multiple books including *Learning CFEngine*, *Learning Hammerspoon* and *Literate Configuration*.
  - Program chair and program committee member for multiple conferences including the RAID symposium, DIMVA conference, the *Computer Security Day* and *Computer Security* conference at UNAM, and others.
  - Member of the Editorial Board of the Computers & Security Journal.

# Experience

## Amazon Web Services

*2021 – Present* *Switzerland*

Senior Global Security Architect *Oct 2021 – Present*

- I work with customers and within AWS to increase security awareness, design and build secure solutions, mentor and develop colleagues and customers in security best practices.
- As Lead Security Architect at the Stellantis *Virtual Engineering Workbench* (VEW) project, I have:
  - Established the VEW security workstream to identify customer security requirements and policies, define and promote security best practices and drive activities related to risk analysis, threat modeling and mitigation definition, prioritization and implementation.
  - Established measurement mechanisms for status and metrics for security activities, which provide visibility to both technical and business stakeholders.
  - Established a *Security Champions* program to promote and transfer security knowledge.
  - Defined and tracked implementation of security features to fulfill both customer business requirements and AWS best practices.

Global Security Architect *Sep 2021 – Sep 2022*

- Worked with AWS global customers to improve security posture and promote secure design and implementation practices.
- In the Volkswagen Digital Production Platform (DPP) program, I established scalable and durable mechanisms to enable DPP partners to work securely in the DPP program. I also created and promoted security learning materials tailored for various roles within the project, and increased security awareness and knowledge by promoting a *Security Guardians* initiative across the organization.

## Swisscom

*2015 – 2021* *Switzerland*

Enterprise Architect and IT Clouds Solution Security Architect *Apr 2019 – Sep 2021*

- As an *Enterprise Architect*, I participated in the design of future products and solutions offered by Swisscom, in collaboration with architects from all other divisions of the company.
- As *Solution Security Architect for Swisscom's Cloud Platforms* —which include *Enterprise Service Cloud*, *Enterprise Application Cloud*, *Dynamic Computing Services*, *Enterprise Cloud for SAP Applications* and related services— I was responsible for the security, compliance and data governance of those services. I defined, prioritized and drove relevant product features and business goals. I also lead the IT Clouds Security Community of Practice and advised engineering teams on compliance, governance and operational activities.
- Selected achievements and activities:
  - Ensured cloud platform and service compliance with internal, contractual and regulatory standards, including ISO27001, ISAE3402/3000 and GDPR.
  - Established and led a community of around 30 *Security Champions* from different teams, who drove security initiatives and promote the security culture within the Swisscom IT Clouds organization.
  - Coordinated threat modeling, audits, penetration tests and security compliance reporting.
  - Coordinated organization- and team-wide processes for risk and vulnerability management.
  - Development of the Swisscom *Platforms* vision for 2025.

Team Lead & Product Owner for Health & State Management *Mar 2016 – Apr 2019*

- I built and led a team which evolved on par with Swisscom cloud platforms to provide their monitoring and logging capabilities. My responsibilities included people management (up to 16 people), definition and prioritization of requirements and roadmaps (in collaboration with Product Managers and other stakeholders), technical architecture, and managing the planning and execution of team activities.
- Selected achievements:
  - Led the transition of the *Enterprise Cloud* LEMM (Logging, Event Management and Monitoring) and Access & Inventory frameworks into maintenance mode as the platform was retired.
  - Defined the scope and mission of the Health and State Management (HSM) team as part of the new *Enterprise Service Cloud* project, and later of other platforms as the *IT Clouds* scope expanded to *Application Cloud*, *Enterprise Cloud for SAP Solutions* and *Dynamic Computing Services*.
  - Defined the logging and monitoring architecture for the *Enterprise Service Cloud* platform based on VMware vRealize Operations and vRealize Log Insight.
  - Led the transition of the *Application Cloud* platform monitoring from the Orchard framework to a TICK-based framework.
  - Defined architecture and oversaw implementation of the Customer Log Forwarding service.
  - Managed business relationship and technical implementation of OpsGenie for alert management in IT Clouds.
- Main technologies involved: VMware vSphere (ESX, vCenter, NSX), VMware vRealize Operations Manager and Log Insight, Ansible (configuration management), OpsGenie (alert management).

Cloud Architect and Orchard Project Lead — *Aug 2015 – Mar 2016*

- Managed a team of three people and led the *Orchard* project through its implementation, production release and further improvements and development.

### Swisscom Cloud Lab — *2014 – 2015* — *U.S.A. (remote)*

Senior Platform Architect — *Aug 2014 – Jul 2015*

- Designed the architecture and implemented the initial prototype for the *Orchard* health-management and self-healing framework for Swisscom's *Application Cloud* Platform-as-a-Service service.
- Main technologies involved: OpenStack (cloud computing infrastructure), Cloud Foundry (application platform), Consul (health management and service discovery), RabbitMQ (message bus), Riemann (event analysis).

### CFEngine AS — *2011 – 2014* — *Norway/U.S.A. (remote)*

Product Manager — *Aug 2013 – Jun 2014*

- Managed the CFEngine language roadmap.
- Created and led the CFEngine Design Center project, which was the foundation for the current CFEngine Build service.
- Coordinated the work on CFEngine third-party integration (e.g. AWS EC2, VMware, Docker and OpenStack).
- Developed code for both the Design Center core and its integrations.

Senior Security Advisor — *Oct 2011 – Jun 2014*

- CFEngine Advocate, with a special focus on security.
- Wrote the book *Learning CFEngine 3*, published by O'Reilly Media, which became the de facto introductory text to CFEngine.
- Gave talks, wrote articles and blog posts, taught classes, and in general spread the word about CFEngine.
- Developed and implemented the strategy for CFEngine as a security component.

### Boundless Innovation and Technology — *2012 – 2014* — *Mexico*

Cofounder, Head of Research and Training — *Jul 2012 – Jul 2014*

- I advised and coordinated teams working on teaching- and security-related products, consulting and services.

### HP Enterprise Services — *2009 – 2011* — *Mexico*

Account Security Officer — *Oct 2010 – Oct 2011*

- Acted as first point of contact for all security-related issues for five HP enterprise customers in Mexico.
- Initiated, advised and managed security-related projects.
- Handled communication and coordination between technical teams involved in security initiatives.
- Involved in all security-related decisions at the sales, design, implementation, delivery and ongoing maintenance stages of IT Outsourcing projects.

IT Outsourcing Service Delivery Consultant — *Nov 2009 – Oct 2010*

- Helped multidisciplinary customer teams (software engineering, IT management, networking, sales and support) by solving complex problems in customer environments.
- Performed analysis, design and implementation of solutions in multiple areas of expertise, including system automation, configuration management, system administration, system design, virtualization, performance and security.

### IBM Zurich Research Lab — *2001 – 2009* — *Switzerland*

Research Staff Member — *Oct 2001 – Oct 2009*

- I was a member of the *Global Security Analysis Laboratory* (GSAL), where I worked in intrusion detection, malware detection and containment, and virtualization security research projects.
- See *Research* for details of my research.

### Sun Microsystems — *1997* — *U.S.A.*

Developer (Intern) — *May 1997 – Aug 1997*

- Developer for the *Bruce* host vulnerability scanner, later released as the Sun Enterprise Network Security Service (SENSS).
- Designed and implemented the first version of the network-based components of *Bruce*, which allowed it to operate on several hosts in a network, controlled from a central location.

**National Autonomous University of Mexico (UNAM)** *1991 – 1996* *Mexico*

HEAD OF COMPUTER SECURITY AREA *Aug 1995 – Aug 1996*

- Founded UNAM's *Computer Security Area*, the University's first team dedicated to computer security, which has evolved into the *Information Security Coordination (UNAM-CERT)*.
- Managed up to nine people working on different projects related to computer security.
- Managed security monitoring for a Cray supercomputer and 22 Unix workstations.
- Provided security services to the whole University, including incident response, security information, auditing and teaching.
- Established the celebration of the *International Computer Security Day* (sponsored by the Association for Computing Machinery) at UNAM. Acted as the main organizer of the event for two years (1994 and 1995). This event has grown and evolved into the *Computer Security Day* and the *Computer Security Congress*.
- Designed and headed development of an audit-analysis tool for Unix systems (SAINT) [24].

SYSTEM ADMINISTRATOR *Nov 1991 – Aug 1995*

- System administrator at UNAM's Supercomputing Center, managing a Cray Y-MP Supercomputer and related systems.
- Managed the Network Queuing Subsystem (NQS),
- Managed and provided support for 22 Unix workstations.
- Monitored the security of the Cray supercomputer and related workstations.
- Other responsibilities: user administration, operating system installation, resource management, security policies.

# Education

**Ph.D. in Computer Science** *West Lafayette, IN, U.S.A.*

PURDUE UNIVERSITY *Aug 1996 – Aug 2001*

- Thesis title: *Using Internal Sensors for Computer Intrusion Detection*.
- Advisor: Eugene H. Spafford.

**M.S. in Computer Science** *West Lafayette, IN, U.S.A.*

PURDUE UNIVERSITY *Aug 1996 – May 1998*

- Advisor: Eugene H. Spafford.

**Bachelor's degree in Computer Engineering** *Mexico City, Mexico*

NATIONAL AUTONOMOUS UNIVERSITY OF MEXICO (UNAM) *Aug 1989 – Jul 1995*

- Thesis title: UNAM/Cray Project for Security in the Unix Operating System (in Spanish, original title: *Proyecto UNAM/Cray de Seguridad en el Sistema Operativo Unix*).

# Certifications




**Certified Information Systems Security Professional (CISSP)** *Since April 2019*

(ISC)², THE INTERNATIONAL INFORMATION SYSTEM SECURITY CERTIFICATION CONSORTIUM

**AWS Certified Solutions Architect – Professional** *Since December 2021*

AMAZON WEB SERVICES TRAINING AND CERTIFICATION

**AWS Certified DevOps Engineer – Professional** *Since May 2022*

AMAZON WEB SERVICES TRAINING AND CERTIFICATION

**AWS Certified Security – Specialty** *Since October 2021*

AMAZON WEB SERVICES TRAINING AND CERTIFICATION

**AWS Certified Data Analytics – Specialty** *Since May 2023*

AMAZON WEB SERVICES TRAINING AND CERTIFICATION

**AWS Certified Solutions Architect – Associate** *Since October 2021*

AMAZON WEB SERVICES TRAINING AND CERTIFICATION

**AWS Certified SysOps Administrator – Associate** *Since February 2022*

AMAZON WEB SERVICES TRAINING AND CERTIFICATION

**AWS Certified Developer – Associate** *Since April 2022*

Amazon Web Services Training and Certification

**Certified SAFe® 5 Agilist** *Since November 2021*

Scaled Agile Inc.

**Certified SAFe® 5 Advanced Scrum Master** *Since March 2022*

Scaled Agile Inc.

# Languages

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|---|---|
| **Spanish** | native |
| **English** | full professional proficiency |
| **German** | intermediate proficiency (B2 level) |

# Research

(see "Publications" for publication reference details)

### Selected research projects at IBM

Phantom *2008 – 2009*

- Security for VMware virtual environments using virtual machine introspection (based on the VMware VMsafe API) to provide intrusion detection and prevention capabilities.
- Publications: [13].

Billy Goat: Active worm detection and capture *2002 – 2008*

- Billy Goat was the first instance of what is today called *honeypots* and *honeynets*.
  - An active worm-detection system, widely deployed (at the time) in the IBM worldwide internal network. It listens for connections to unused IP address ranges and actively responds to those connections to accurately detect worm-infected machines, and in many cases capture the worms themselves. Billy Goat is engineered for distributed deployment, with each device containing standalone detection and reporting capabilities, together with data centralization features that allow network-wide data analysis and reporting.
- Publications: [18, 25]

Router-based Billy Goat *2005 – 2007*

- An active worm-capture device deployed at the network boundary and coupled with the border router, that allows the Billy Goat to effectively and automatically spoof every unused IP address outside the local network. This makes it possible for the Router-based Billy Goat to accurately detect local infected machines and prevent them from establishing connections to the outside, limiting the propagation of the worms to the outside network.
- Publications: [16]

SOC in a Box *2005 – 2007*

- Integrated device containing multiple security tools: intrusion detection, worm detection, vulnerability scanning and network discovery. Precursor to what is today called *Unified Threat Management* systems.

Exorcist *2001 – 2002*

- Host-based, behavior-based intrusion detection using sequences of system calls.

### Ph.D. Thesis Research

Using internal sensors and embedded detectors for intrusion detection

- Study of data collection methods for intrusion detection systems.
- Implementation of novel methods for data collection in intrusion detection systems.
- Analysis of the properties, advantages and disadvantages of internal sensors and embedded detectors as data collection and analysis elements in intrusion detection systems.
- Publications: [11, 19, 20, 27, 32]

### Additional research projects

#### Using autonomous agents for intrusion detection

- Design and documentation of an architecture (AAFID) to perform distributed monitoring and intrusion detection using autonomous agents.
- Implementation of a prototype according to the architecture. This prototype is published as open source.
- Exploration of research issues in the distributed intrusion detection area.
- Publications: [21, 22, 28, 35, 33, 34].

#### Analysis of a denial-of-service attack on TCP/IP (Synkill)

- Collaborated in the analysis of the SYN-flooding denial-of-service attack against TCP and in the implementation of a defense tool.
- Publications: [23], awarded the 2020 IEEE Security & Privacy Test of Time Award.

## Software Development Projects

**Publicly available software projects: see https://github.com/zzamboni/ and https://gitlab.com/zzamboni**

### Other software projects (not publicly available)

Pilatus (IBM) *2005 – 2007*

A system installer that allows arbitrary system installation and configurations, allowing for both proprietary and open source components to be installed in an automated fashion. Open source components can be downloaded directly from their original source to avoid distributing them.

SOC in a Box (IBM) *2005 – 2007*

A specialized Linux distribution containing multiple security services for integrated security monitoring in small and medium networks. Implementation includes also backend infrastructure components for system installation, configuration and upgrade; and data centralization, analysis and reporting.

Billy Goat (IBM) *2002 – 2007*

A specialized Linux distribution containing multiple sensors for detection of large-scale automated attacks. Implementation includes also backend infrastructure components for system configuration and upgrade, data centralization, analysis and reporting.

Embedded Sensors Project (Purdue University) *1999 – 2001*

A system of sensors for intrusion detection developed in OpenBSD through code instrumentation. Developed as part of my Ph.D. thesis work.

## Honors & Awards

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|---|---|---|
| May 2020 | **IEEE Security & Privacy Test of Time Award (IEEE S&P page, CERIAS blog post),** IEEE | *U.S.A.* |
| 2010 | **CFEngine Champion,** CFEngine AS | *Norway* |
| Jul 2001 | **Josef Raviv Memorial Postdoctoral Fellowship,** IBM | *U.S.A.* |
| Apr 2001 | **Member of Phi Beta Delta,** honor society recognizing scholarly achievement | *U.S.A.* |
| Sep 2000 | **UPE Microsoft Scholarship Award,** honor society recognizing scholarly achievement | *U.S.A.* |
| Apr 1998 | **Member of Upsilon Pi Epsilon,** the ACM Computer Sciences honor society | *U.S.A.* |
| May 1996 | **Fulbright Scholarship,** for pursuing Ph.D. studies at Purdue University | *Mexico* |

# Program Committees and Boards

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|---|---|---|
| 2011–2013 | **Editorial Board Member**, Computers & Security Journal | |
| 2007–2017 | **Steering Committee Member**, Intl. Symposium on Recent Advances in Intrusion Detection | |
| 2006 | **Program Chair**, 9th Intl. Symposium on Recent Advances in Intrusion Detection (RAID) | *Germany* |
| 2001–2005 | **Program Committee Member**, Intl. Symposium on Recent Advances in Intrusion Detection | |
| 2009 | **Program Co-chair**, IBM Academy of Technology Security and Privacy Symposium | |
| 2009 | **Program Chair**, ZISC Workshop on Security in Virtualized Environments and Cloud Computing | *Switzerland* |
| 2008 | **Program Chair**, Detection of Intrusions and Malware & Vulnerability Assessment (DIMVA) | *France* |
| 2007 | **Program Committee Member**, IEEE Security and Privacy Symposium | *U.S.A.* |
| 2003–2007 | **Program Committee Member**, Annual Computer Security Applications Conference (ACSAC) | |
| 1994–2000 | **Program Committee Member**, Computer Security Day Conference | *Mexico* |
| 1994–1995 | **Founder and organizer**, Computer Security Day Conference | *Mexico* |

# Teaching and Advising

## Students

Urko Zurutuza Ortega, Mondragon University, Spain — *Ph.D. co-advisor* — *2005 – 2008*

- Thesis: Data Mining Approaches for Analysis of Worm Activity Towards Automatic Signature Generation

Milton Yates, ENST Bretagne, France — *External Diploma Thesis advisor* — *2005*

- Thesis: The Router-based Billy Goat Project

Candid Wüest, ETH Zurich, Switzerland — *Diploma Thesis advisor* — *2002 – 2003*

- Thesis: Desktop Firewalls and Intrusion Detection

Daniele Sgandurra, University of Pisa, Italy — *Internship advisor* — *2009*

- Project: Design and implementation of process injection using virtual machine introspection.

Martin Carbone, Georgia Institute of Technology, U.S.A. — *Internship advisor* — *2007*

Project: Implementation of a proof of concept Hyperjacking attack on Intel platform.

# Other Professional Activities

| | | |
|---|---|---|
| 1998– | **The Association for Computing Machinery (ACM)**, Member | |
| 2000 | **Purdue.pm, the Purdue Perl Users Group**, Founder | *U.S.A.* |
| 1999 | **Purdue University Chapter of Upsilon Pi Epsilon**, President | *U.S.A.* |
| 1998 | **Purdue University Chapter of Upsilon Pi Epsilon**, Secretary | *U.S.A.* |

# Publications, Talks and Intellectual Property

## Books


[1] Diego Zamboni. *Publishing with Emacs, Org-mode and Leanpub*. June 2020. URL: https://leanpub.com/emacs-org-leanpub.

[2] Diego Zamboni. *Literate Configuration*. Nov. 2019. URL: https://leanpub.com/lit-config.

[3] Diego Zamboni. *Utilerías de Unix (Unix utilities course notes)*. Aug. 2019. URL: https://leanpub.com/utileras-unix.

[4] Diego Zamboni. *Learning Hammerspoon*. Self published, Oct. 2018. URL: https://leanpub.com/learning-hammerspoon.

[5] Diego Zamboni. *Learning CFEngine*. O'Reilly Media, Inc. 2012–2017, afterwards self-published, 2017. ISBN: 9781449312206. URL: http://cf-learn.info/.


## Editorial Activities


[6] *Computers & Security Journal*. Elsevier. Member of the Editorial Board. 2011–2013.

[7] Deborah Frincke, Andreas Wespi, and Diego Zamboni, eds. *Computer Networks* 51.5 (Apr. 2007): *From Intrusion Detection to Self-Protection*. ISSN: 1389-1286. URL: http://dx.doi.org/10.1016/j.comnet.2006.10.004.

[8] Diego Zamboni and Christopher Kruegel, eds. Recent Advances in Intrusion Detection (RAID): 9th International Symposium (Hamburg, Germany, Sept. 20–22, 2006). Lecture Notes in Computer Science. Secaucus, NJ, USA: Springer-Verlag New York, Inc., 2006. ISBN: 354039723X.

[9] Alfonso Valdes and Diego Zamboni, eds. Recent Advances in Intrusion Detection (RAID): 8th International Symposium (Seattle, WA, U.S.A. Sept. 7–9, 2005). Lecture Notes in Computer Science. Secaucus, NJ, USA: Springer-Verlag New York, Inc., 2005. ISBN: 3540317783.

[10] Diego Zamboni, ed. *Software: Practice and Experience* 33.5 (Apr. 2003): *Special issue on "Security Software"*. URL: http://onlinelibrary.wiley.com/doi/10.1002/spe.v33:5/issuetoc.

### Theses

[11] Diego Zamboni. "Using Internal Sensors for Computer Intrusion Detection". CERIAS TR 2001-42. PhD thesis. West Lafayette, IN: Purdue University, Aug. 2001. URL: https://www.cerias.purdue.edu/apps/reports_and_papers/view/1800.

[12] Diego Zamboni. "Proyecto UNAM/Cray de Seguridad en el Sistema Operativo Unix (*UNAM/Cray project for Unix System Security*)". Spanish. B.Sc. Thesis. Universidad Nacional Autonoma de México, June 1995. URL: https://zzamboni.org/files/theses/zamboni-bachelors-thesis.pdf.

### Refereed Papers

[13] Mihai Christodorescu et al. "Cloud Security is Not (Just) Virtualization Security: A Short Paper". In: *Proceedings of the 2009 ACM Workshop on Cloud Computing Security*. Chicago, Illinois, USA: ACM, 2009, pp. 97–102. ISBN: 978-1-60558-784-4. DOI: 10.1145/1655008.1655022.

[14] U. Zurutuza et al. "Un marco inteligente para el análisis de tráfico generado por gusanos en Internet (An intelligent framework for analysis of worm-generated Internet traffic)". Spanish. In: *Actas de la X Reunión Española sobre Criptología y Seguridad de la Información (X Spanish Meeting on Cryptology and Information Security)*. Sept. 2008.

[15] Urko Zurutuza, Roberto Uribeetxeberria, and Diego Zamboni. "A data mining approach for analysis of worm activity through automatic signature generation". In: *Proceedings of the 1st ACM workshop on AISec (AISec'08)*. Alexandria, Virginia, USA: Association for Computing Machinery, Oct. 2008, pp. 61–70. ISBN: 978-1-60558-291-7. URL: http://doi.acm.org/10.1145/1456377.1456394.

[16] Diego Zamboni, James Riordan, and Milton Yates. "Boundary detection and containment of local worm infections". In: *Proceedings of the 3rd Workshop on Steps to Reducing Unwanted Traffic on the Internet (SRUTI'07)*. Usenix. June 2007. URL: http://www.usenix.org/events/sruti07/tech/full_papers/zamboni/zamboni.pdf.

[17] Urko Zurutuza, Roberto Uribetxeberria, and Diego Zamboni. "Anàlisis de datos procedentes de un sistema de detección de gusanos mediante técnicas de clustering (Analysis of data from a worm-detection system using clustering techniques)". In: *Actas del II Simposio sobre Seguridad Informática (SSI'2007), II Congreso Español de Informática (CEDI 2007) (Proceedings of the II Symposium on Computer Security, II Spanish Conference on Informatics)*. Sept. 2007, pp. 87–94.

[18] James Riordan, Diego Zamboni, and Yann Duponchel. "Building and deploying Billy Goat, a worm-detection system". In: *Proceedings of the 18th Annual FIRST Conference*. June 2006.

[19] Florian Kerschbaum, Eugene H. Spafford, and Diego Zamboni. "Using internal sensors and embedded detectors for intrusion detection". In: *Journal of Computer Security* 10.1,2 (2002), pp. 23–70. URL: https://www.researchgate.net/publication/220065478_Using_Internal_Sensors_and_Embedded_Detectors_for_Intrusion_Detection.

[20] Florian Kerschbaum, Eugene H. Spafford, and Diego Zamboni. "Using embedded sensors for detecting network attacks". In: *Proceedings of the 1st ACM Workshop on Intrusion Detection Systems*. Ed. by Deborah Frincke and Dimitris Gritzalis. CERIAS TR 2000-25. ACM SIGSAC. Nov. 2000. URL: https://www.cerias.purdue.edu/apps/reports_and_papers/view/1641/.

[21] Eugene H. Spafford and Diego Zamboni. "Intrusion Detection using Autonomous Agents". In: *Computer Networks* 34.4 (Oct. 2000), pp. 547–570. URL: http://dx.doi.org/10.1016/S1389-1286(00)00136-5.

[22] Jai Sundar Balasubramaniyan et al. "An Architecture for Intrusion Detection using Autonomous Agents". In: *Proceedings of the Fourteenth Annual Computer Security Applications Conference*. IEEE Computer Society, Dec. 1998, pp. 13–24. URL: http://zzamboni.org/files/pubs/aafid-acsac98.pdf.

[23] Christoph L. Schuba et al. "Analysis of a Denial of Service Attack on TCP". In: *Proceedings of the 1997 IEEE Symposium on Security and Privacy*. Awarded the 2020 IEEE Security & Privacy Test of Time Award. IEEE Computer Society. IEEE Computer Society Press, May 1997, pp. 208–223. URL: https://www.cerias.purdue.edu/apps/reports_and_papers/view/605.

[24] Diego Zamboni. "SAINT —A Security Analysis Integration Tool". In: *Proceedings of the 1996 Systems Administration, Networking and Security Conference*. Washington, D.C., May 1996. URL: http://zzamboni.org/files/pubs/SAINT.pdf.

### Tech Reports

[25] James Riordan, Diego Zamboni, and Yann Duponchel. *Billy Goat, an Accurate Worm-Detection System*. Research Report RZ 3609. IBM Research, Nov. 2005. URL: https://dominoweb.draco.res.ibm.com/reports/rz3609_revised.pdf.

[26] Eugene Spafford and Diego Zamboni. *Data Collection mechanisms for intrusion detection systems*. CERIAS Technical Report 2000-08. 1315 Recitation Building, West Lafayette, IN: CERIAS, Purdue University, June 2000. URL: https://www.cerias.purdue.edu/apps/reports_and_papers/view/1842.

[27] Diego Zamboni. *Doing intrusion detection using embedded sensors— Thesis proposal*. CERIAS Technical Report 2000-21. West Lafayette, IN: CERIAS, Purdue University, Oct. 2000. URL: https://www.cerias.purdue.edu/apps/reports_and_papers/view/581.

[28] Jai Sundar Balasubramaniyan et al. *An Architecture for Intrusion Detection using Autonomous Agents*. Technical Report 98-05. COAST Laboratory, Purdue University, May 1998. URL: https://www.cerias.purdue.edu/apps/reports_and_papers/view/65.

### Presentations at Conferences and Workshops

[29] Diego Zamboni and Bill Chapman. *Chaos Heidi vs. Orchard: Self-Disruption and Healing in a Cloud Foundry-Based Service Environment*. Presented at the Cloud Foundry Summit Silicon Valley 2016. Presentation: https://www.youtube.com/watch?v=Wr4E--kr_KE. May 2016. URL: http://sched.co/6aQ2.

[30] Diego Zamboni and Mark Burgess. *The Future of In-Container Configuration Management*. Invited talk at the 2014 Usenix Configuration Management Summit (UCMS'14). June 2014. URL: https://www.usenix.org/conference/ucms14/summit-program/presentation/zamboni.

[31] Mike Svoboda and Diego Zamboni. *Leveraging In-Memory Key Value Stores for Large-Scale Operations*. Invited talk at the 27th Large Installation System Administration (LISA) Conference. Nov. 2013. URL: https://www.usenix.org/conference/lisa13/leveraging-memory-key-value-stores-large-scale-operations.

[32] Eugene H. Spafford and Diego Zamboni. *Design and implementation issues for embedded sensors in intrusion detection*. Presented at the Third International Workshop on Recent Advances in Intrusion Detection (RAID2000). Oct. 2000. URL: http://zzamboni.org/files/pubs/sensors-raid2000.pdf.

[33] Diego Zamboni. *Building a Distributed Intrusion Detection System with Perl*. Presented at The Perl Conference 4.0. Monterey, CA, July 2000. URL: http://zzamboni.org/files/pubs/tpc40.pdf.

[34] Eugene H. Spafford and Diego Zamboni. "New directions for the AAFID architecture". In: *Proceedings of the Second International Workshop on Recent Advances in Intrusion Detection (RAID99)*. Online proceedings, available at http://www.raid-symposium.org/raid99/. West Lafayette, IN, Sept. 1999. URL: https://www.cerias.purdue.edu/apps/reports_and_papers/view/3487.

[35] Eugene H. Spafford and Diego Zamboni. "AAFID: Autonomous Agents for Intrusion Detection". In: *Proceedings of the First International Workshop on Recent Advances in Intrusion Detection (RAID98)*. Online proceedings, available at http://www.raid-symposium.org/raid98/. Louvain-la-Neuve, Belgium, Sept. 1998.

### Invited Talks and Articles

[36] Mark Burgess and Diego Zamboni. "CFEngine's Decentralized Approach to Configuration Management". In: *InfoQ* (June 2014). URL: http://www.infoq.com/articles/cfengine-view-on-it-automation.

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[39] Diego Zamboni. *From Intrusion Detection to Remediation and Beyond: Evolution, Trends, and Research at IBM*. Invited talk at the annual meeting of the Swiss Chapter of the Sigma XI Honorary Scientific Society. Nov. 2006.

[40] James Riordan, Andreas Wespi, and Diego Zamboni. "How to Hook Worms". In: *IEEE Spectrum* (May 2005). URL: https://spectrum.ieee.org/computing/networks/how-to-hook-worms.

[41] Diego Zamboni. *Intrusion what? From detection to prevention and beyond*. Talk at the Zurich Information Security Center Information Security Colloquium. Dec. 2005.

[42] James Riordan and Diego Zamboni. "Billy Goat Detects Worms and Viruses". In: *ERCIM News* 56 (Jan. 2004). URL: http://www.ercim.org/publication/Ercim_News/enw56/riordan.html.

[43] Diego Zamboni. Diez Años de Aciertos y Fallas — ¿Qué Hemos Aprendido y Qué nos Depara el Futuro en la Seguridad? *(Ten years of hits and misses — what have we learned, and what does the future in security hold for us?)* Keynote talk, presented at the 2004 Computer Security Congress in Mexico City. May 2004.

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